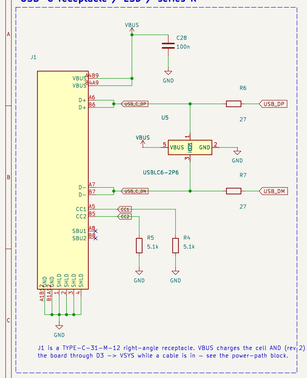
simple_earring — generated schematic

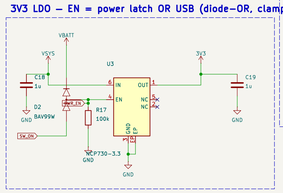
157 components / 97 nets / 4-layer JLCPCB stackup. Power rails (GND, 3V3, VBATT, VSYS, VLED, VBUS, VDD_CORE) are power symbols; only cross-block signals use global labels. Generated from scripts/circuit.py — do not hand-edit; rebuild with `python3 build.py --only schematic`.

USB-C receptacle / ESD / series R

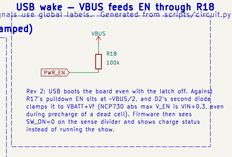


J1 is a TYPE-C-31-M-12 right-angle receptacle. VBUS charges the cell AND (rev 2) runs the board through D3 -> VSYS while a cable is in — see the power-path block.

3V3 LDO — EN = power latch OR USB (diode-OR, clamped)

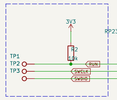


USB wake — VBUS feeds EN through R18

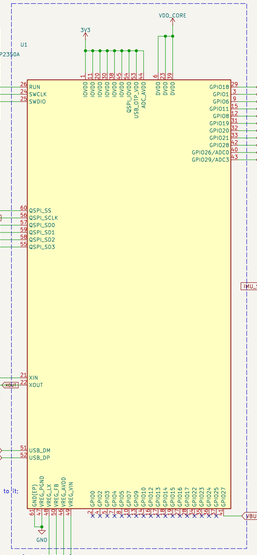


Rev 2: USB boots the board even with the latch off. Against R17's pulldown EN sits at ~VBUS/2, and D2's second diode clamps it to VBATT+Vf (MCP730 abs max V_EN is VIN+0.3, even during precharge of a dead cell). Firmware then sees SW_ON=0 on the sense divider and shows charge status instead of running the show.

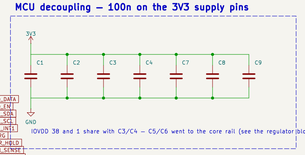
RUN / SWD



RP2350A

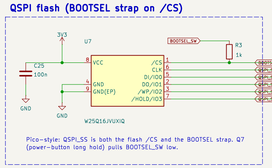


MCU decoupling — 100n on the 3V3 supply pins



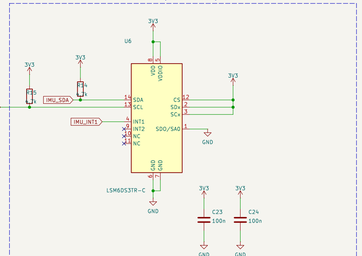
IOVDD 3B and 1 share with C3/C4 — C5/C6 went to the core rail (see the regulator block)

QSPI flash (BOOTSEL strap on /CS)



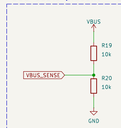
Pico-style: QSPI_SS is both the flash /CS and the BOOTSEL strap. Q7 (power-button long hold) pulls BOOTSEL_SW low.

IMU — LSM6DS3TR-C on I2C1 (addr 0x6A)



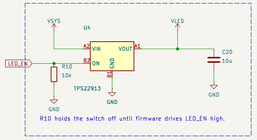
CS/SDx/SCx -> VDDIO selects I2C mode 1 and idles the aux master; SDO/SA0 -> GND sets address 0x6A.

USB cable present — 10k:10k tap on VBUS -> ADC1



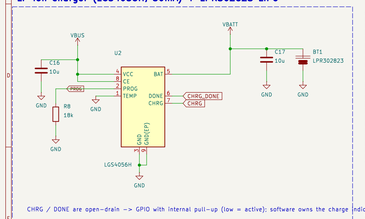
CHRG and DONE both release at termination, so they cannot tell 'unplugged' from 'plugged in and full'. 10k:10k, not the 100k/1M used elsewhere: PWR_EN backfeeds through R18 and would float this node high, so the tap has to be stiff enough to swamp it (0.33V no cable, 2.2–2.75V with one). That is also what holds Q2's gate low enough to keep the battery path on.

VLED high-side load switch (LEDs draw zero when off)



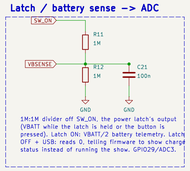
R10 holds the switch off until firmware drives LED_EN high.

Li-Ion charger (LGS4056H, 50mA) + LPR302823 LiPo



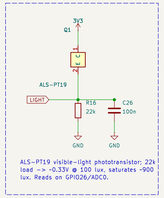
CHRG / DONE are open-drain -> GPIO with internal pull-up (low = active); software owns the charge indicator.

Latch / battery sense -> ADC



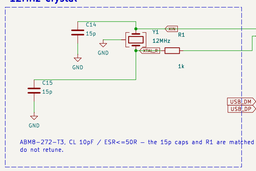
1M:1M divider off SW_ON, the power latch's output (VBATT while the latch is held or the button is pressed). Latch ON: VBATT/2 battery telemetry. Latch OFF + USB: reads 0, telling firmware to show charge status instead of running the show. GPIO29/ADC3.

Ambient light sensor -> ADC0 (rev 2)



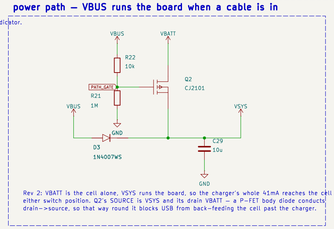
ALS-PT19 visible-light phototransistor: 22k load -> ~0.33V @ 100 lux, saturates ~900 lux. Reads on GPIO26/ADC0.

12MHz crystal



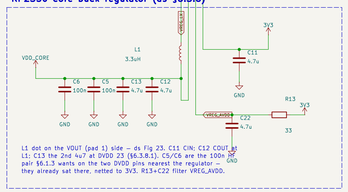
ABM8-272-T3, CL 10pF / ESR<=50R — the 15p caps and R1 are matched to it; do not retune.

power path — VBUS runs the board when a cable is in



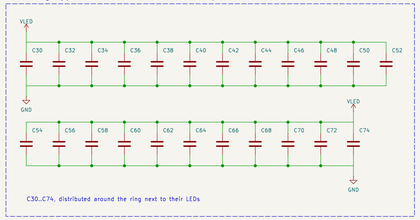
Rev 2: VBATT is the cell alone, VSYS runs the board, so the charger's whole 41mA reaches the cell in either switch position. Q2's SOURCE is VSYS and its drain VBATT — a P-FET body diode conducts drain->source, so that way round it blocks USB from back-feeding the cell past the charger.

RP2350 core buck regulator (ds §6.3.8)



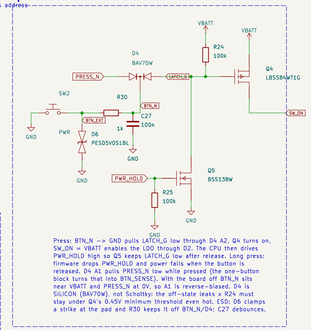
L1 dot on the VOUT (pad 1) side — ds Fig 23. C11 CIN: C12 COUT at L1: C13 the 2nd 4u7 at DVDD 23 (§6.3.8.1). C5/C6 are the 100n pair §6.1.3 wants on the two DVDD pins nearest the regulator — they already sat there, netted to 3V3. R13+C22 filter VREG_AVDD.

LED ring bypass — 100n VLED–GND, one per two LEDs



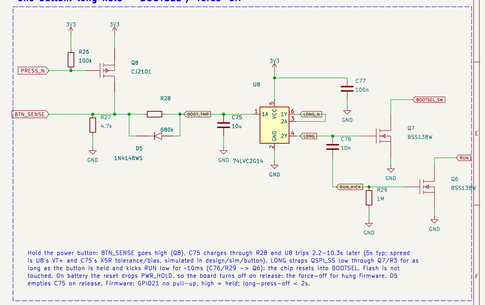
C30..C74, distributed around the ring next to their LEDs

power button + soft latch



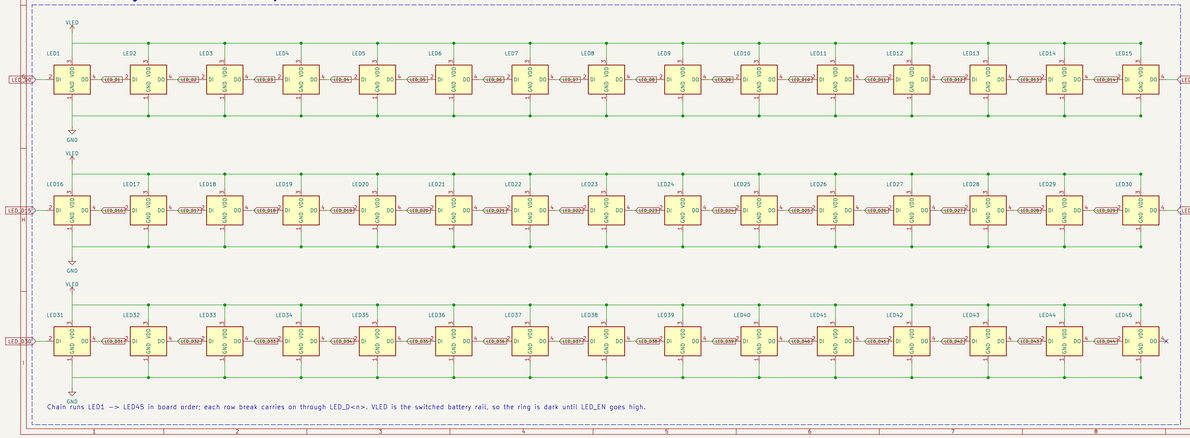
Press: BTN_N -> GND pulls LATCH_G low through D4 A2, Q4 turns on, SW_ON -> VBATT enables the LDO through D2. The CPU then drives PWR_HOLD high so Q5 keeps LATCH_G low after release. Long press: firmware drops PWR_HOLD and power fails when the button is released. D4 A1 pulls PRESS_N low while pressed (the one-button block turns that into BTN_SENSE), With the board off BTN_N sits near VBATT and PRESS_N at 0V, so A1 is reverse-biased. D4 is SILICON (BAV70W), not Schottky: the off-state leaks x R24 must stay under Q4's 0.45V minimum threshold even hot. ESD: D6 clamps a strike at the pad and R30 keeps it off BTN_N/D4; C27 debounces.

one button: long hold = BOOTSEL / force-off



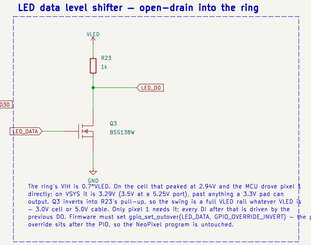
Hold the power button: BTN_SENSE goes high (Q8). C75 charges through R28 and U8 trips 2.2–10.3s later (5s typ; spread is U8's VT+ and C75's X5R tolerance/bias, simulated in design/sim/button). LONG straps QSPI_SS low through Q7/R3 for as long as the button is held and kicks RUN low for ~10ms (C76/R29 -> Q6); the chip resets into BOOTSEL. Flash is not touched. On battery the reset drops PWR_HOLD, so the board turns off on release; the force-off for hung firmware. D5 empties C75 on release. Firmware: GPIO21 no pull-up, high = held; long-press-off < 2s.

Addressable LED ring — 45 x SK6805-EC10 daisy chain



Chain runs LED1 -> LED45 in board order; each row break carries on through LED_D<n>. VLED is the switched battery rail, so the ring is dark until LED_EN goes high.

LED data level shifter — open-drain into the ring



The ring's VIH is 0.7*VLED. On the cell that peaked at 2.94V and the MCU drove pixel 1 directly: on VSYS it is 3.29V (3.5V at a 5.25V port), past anything a 3.3V pad can output. Q3 inverts into R23's pull-up, so the swing is a full VLED rail whatever VLED is — 3.0V cell or 5.0V cable. Only pixel 1 needs it: every DI after that is driven by the previous DO. Firmware must set gpio_set_outover(LED_DATA, GPIO_OVERRIDE_INVERT) — the pad override sits after the PIO, so the NeoPixel program is untouched.